Close pairs and blending of SZ clusters on a Poisson halo sky

Analytic results checked against full-sky simulations code & full report: `sz-poisson-blending/` 10 November 2026

1 Set-up

- **Halos.** Tinker08 HMF (M_{500c}), Planck18, $10^{13} < M_{500}/M_\odot < 10^{16}$, $0.01 < z < 3$. Positions: a Poisson process on the full sky (no clustering).
- **Signal.** Planck 2013 Y_{500}–M_{500}, $(1-b) = 0.8$, $\sigma_{\ln Y} = 0.173$; Arnaud GNFW profile cut at $5R_{500}$.
- **Instrument.** Beam FWHM $1.4'$; white noise 4×10^{-6} (y-arcmin); no CMB or foregrounds.
- **Detection.** Matched filter at the cluster's own θ_{500}. $q = Y_{500}/\sigma_Y(\theta_{500})$, $q_{\rm obs} = q + \mathcal{N}(0,1)$, cut at $q_{\rm cut} = 4$–7.
- **Counts.** 2.93×10^7 halos above 10^{13}, 3.96×10^5 above 10^{14}. Detections at $q_{\rm cut} = 4, 5, 6, 7$: 132 125, 85 912, 61 630, 46 575, i.e. $\Sigma_{\rm det} = 3.20$, 2.08, 1.49, 1.13 deg^{-2}.

2 Analytic results

Q1: N clusters in an aperture. Selecting by mass, $q_{\rm obs}$ or z thins a Poisson process, and the result is still Poisson. For surface density Σ and a cap of radius θ:

$$\mu = \Sigma\, 2\pi(1-\cos\theta) \simeq \Sigma\pi\theta^2, \qquad P(N=n) = e^{-\mu}\mu^n/n!, \qquad P(\geq k) = 1 - \textstyle\sum_{j<k} e^{-\mu}\mu^j/j!\,.$$

- **Centred on a detected cluster.** The *other* objects follow the same Pois(μ) (Slivnyak–Mecke). This also holds for neighbours from another population B, with $\mu_B = \Sigma_B A(\theta)$, whatever the centre's z or q. Do not count the centre.
- **k-th neighbour.** $P(d_k \leq \theta) = \gamma(k,\mu)/\Gamma(k)$.
- **Whole sky.**
  - Pairs closer than θ: $E = \bar{N}\mu/2$, Var $= E(1+2\mu)$.
  - Triples with all separations $< \theta$: $\bar{N}\mu^2 p_3/6$, $p_3 = 1 - 3\sqrt{3}/4\pi$.
  - Chance that at least one such configuration exists: $1 - e^{-E}$.
- **Flat sky.** $\pi\theta^2$ is off by a factor $\simeq 1 + \theta^2/12$, which only matters above about $5°$.

Q2: blending at a threshold. Neighbour B adds $x_B = \rho(\theta_A, \theta_B, s_{AB})\, Y_B/\sigma_Y(\theta_A)$ to the SNR of primary A. Here ρ is the filter's cross-response, with $\rho = 1$ for a source at the centre. The blended SNR is $q_b = q_A + n + \delta - \mu_{\bar{y}}$, with $\delta = \sum_B x_B$. Four outcomes are counted:

- **B1, merged pair:** two detectable clusters within θ_b. $f_{\rm B1} = 1 - e^{-\Sigma_{\rm det} A(\theta_b)}$.
- **B2, contamination:** Y over-estimated by more than ε.
- **B3, boosting:** detected only thanks to neighbours.
- **Lost:** detectable alone but not detected, either absorbed into a brighter peak or pushed below threshold.

The summed neighbour signal is *compound Poisson* (the "P(D)" of confusion theory):

$$\phi_\delta(k) = \exp\Big[\int dx\, R(x)\left(e^{ikx} - 1 - ikx\right)\Big], \qquad R(x) = \int dz\, d\ln M\, d\ln Y\, \frac{d^3N}{d\Omega\, dz\, d\ln M}\, p(\ln Y) \int d^2s\, \delta_D\big(x - \rho Y/\sigma_A\big).$$

The $-ikx$ term removes the map mean. By Slivnyak's theorem the same law holds around a halo as around a random point. Fractions are taken *over the selected sample*, and A must be the brightest contributor:

$$f_{\rm B2} = \frac{\int dN_A\, e^{-\Lambda(x>q_A)} \int d\delta\, p(\delta)\, \mathbf{1}[\delta - \mu_{\bar{y}} > \varepsilon q_A]\, \Phi(q_A + \delta - \mu_{\bar{y}} - q_{\rm cut})}{\int dN_A\, e^{-\Lambda(x>q_A)} \int d\delta\, p(\delta)\, \Phi(q_A + \delta - \mu_{\bar{y}} - q_{\rm cut})}\,,$$

where $\Lambda(x > q_A) = \int_{q_A}^{\infty} R(x)\, dx$ is the expected number of neighbours brighter than A. B3 and Lost follow from the same integrand.

- **The filter sees the mean y.** A white-noise matched filter is not compensated, so it responds to the mean $\bar{y}$ through $\mu_{\bar{y}} = \bar{y}\,\sigma_A\,{\rm RATIO}/N_w$, where RATIO is the filter's response to a uniform y background and N_w the white-noise level. With $\bar{y} = 6.4 \times 10^{-7}$ this is a shift of 0.4σ ($0.5'$ filter) to 3σ ($10'$). Without mean removal, blending fractions roughly triple and do not converge as the mass floor is lowered.
- **Selection.** A detection is more likely to pass the cut when its neighbours add signal. So formulas that ignore selection, such as $1 - e^{-\lambda}$, underestimate contamination by $\times 1.3$–1.5.

3 Simulations

- **Q1.** 120 full-sky catalogues, plus 2×1000 skies with a mass floor and 20 000 skies of rare objects. Neighbours counted with KD-trees on the sphere for $0.5'$–$60'$ and $N = 0$–6.
- **Q2, catalogue level.** The P(D) compared with direct sums over 60 skies.
- **Q2, map level.** 324 flat patches of $10° \times 10°$ (32 400 deg^2). Every halo above $10^{13} M_\odot$ painted, noise added, map mean removed. Then a 12-scale matched filter ($0.5'$–$10'$), peaks taken at the maximum SNR over scales and merged within $2'$. Each peak is split exactly into per-halo contributions.
- **Checks.** Every number was confirmed by two independent re-implementations or re-run on fresh seeds.

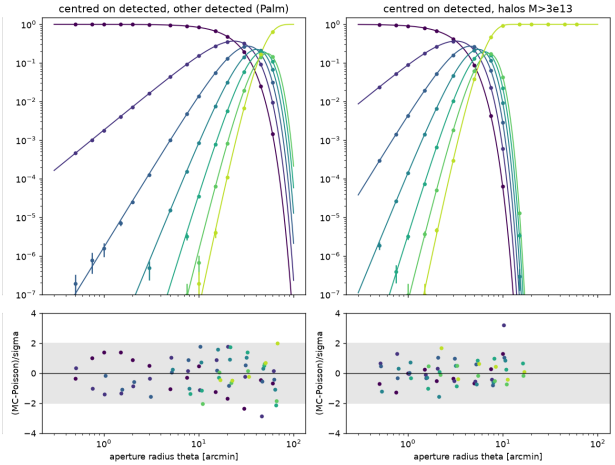


Figure 1: $P(N = n)$ of *other* objects within θ of a detected ($q > 5$) cluster, over 120 full skies. Left: other detections. Right: halos with $M > 3 \times 10^{13}$. Lines are Pois(μ) with no free parameter; colours run from $n = 0$ (dark) to $n \geq 6$ (yellow). Bottom panels: residuals in σ. A sky bootstrap gives a global $p = 0.40$.

$P(\geq 1$ neighbour), $q > 5$ centre	$1'$	$2'$	$5'$	$10'$
other detection ($q > 5$)	0.18%	0.72%	4.4%	16.6%
halo $M > 10^{14}$	0.84%	3.3%	18.9%	56.8%
halo $M > 3 \times 10^{13}$	9.3%	32%	91%	99.99%
$P(\geq 2)$, other detections	$2{\times}10^{-6}$	$3{\times}10^{-5}$	0.10%	1.5%
pairs on the full sky	78	312	1952	7807

Monte Carlo agrees to rms 0.8σ over 72 entries. MC triples at $5'$: 17.2 ± 0.4 (analytic 17.3).

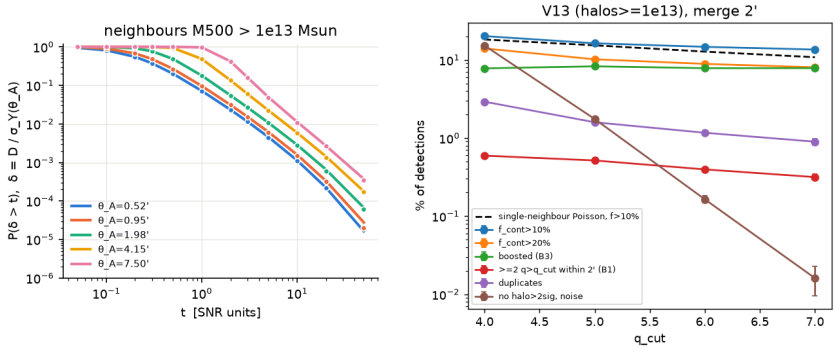


Figure 2: Left: probability that the summed neighbour signal in a filter of scale θ_A exceeds t (in units of the noise σ). Lines: compound-Poisson FFT. Points: direct sums over 60 skies (rms $z = 0.93$ in the tail). **Right:** map-level blending versus $q_{\rm cut}$, as a fraction of *all* map peaks. Projected neighbours (blue, orange, green) dominate. Merged pairs of detectable clusters (red) stay below 0.6%. Pure-noise peaks (brown) matter only at $q_{\rm cut} = 4$.

Table 1: Blending versus $q_{\rm cut}$. Cat.: analytic compound-Poisson fractions of blended detections; Monte Carlo agrees to rms $z = 0.98$ over 363 tests. Map: fractions of real map detections (the denominator excludes noise peaks, unlike Fig. 2), $\pm$0.1–0.2%. Lost: percentage of halos detectable alone that are not detected.

	B1 (map)	B2: Y bias > 10%		B2: Y bias > 20%		B3: boosted		Lost	
$q_{\rm cut}$	≥ 2 det. in $2'$	cat.	map	cat.	map	cat.	map	cat.	map
4	0.59%	18.5%	13.7%	12.4%	7.8%	8.7%	7.8%	10.9%	11.8%
5	0.52%	16.2%	14.5%	10.5%	8.4%	8.1%	8.3%	9.0%	9.7%
6	0.40%	14.5%	13.8%	9.1%	8.1%	7.6%	7.8%	7.9%	8.5%
7	0.32%	13.1%	12.9%	8.1%	7.4%	7.1%	7.5–7.9%	7.1%	7.5%

4 Robustness and beyond Poisson

- **Instrument and model.** Across 11 noise, beam, $z_{\rm max}$ and HMF variants, Q1 stays Poisson ($\chi^2 = 124/124$); only $\Sigma_{\rm det}$ changes. Contamination and boosting scale by $\times 1.3$ for noise $\times 0.5$, $\times 1.4$ for a $5'$ beam, and $\times 0.95$ for a different HMF. Y-scatter has no effect.
- **Mass floor.** With the mean removed, a floor of 10^{13} is converged. Halos above 10^{14} supply $\sim$83% of B2 and B3.
- **Clustering** (Limber, Tinker10 bias) multiplies close-pair probabilities by $1 + \bar{w} = 1.13$–1.27 at 1–$5'$ and blending fractions by $\times 1.08$–1.13.
- **Halo exclusion** is negligible: about 0.1% of detected close pairs overlap in 3D. Blending is projection along the line of sight.
- **The detected catalogue is not Poisson.** Neighbours boost each other over the threshold, so close pairs at 2–$5'$ are 1.5–2.7$\times$ the Poisson rate. Pairs closer than about 0.35 FWHM are suppressed.

5 Conclusions

- **Q1 is purely geometric.** $P(N) = {\rm Pois}(\Sigma A)$ for random or cluster-centred apertures and for any selection. Simulations confirm it to $\lesssim 1\sigma$. For an SO-like survey at $q > 5$, 0.7% of clusters have another detection within $2'$, while a third have a $> 3 \times 10^{13}$ halo there.
- **Q2 is dominated by confusion, not by pairs.** Merged pairs of detectable clusters are only about 0.5% of detections. About 15% of $q > 5$ detections have Y biased by more than 10% by mostly undetectable halos at other redshifts. About 8% exist only because of that boost, and about 10% of detectable halos are lost.
- **No single blending radius describes contamination.** The radius needed to reproduce B2 grows from $8'$ to $12'$ with $q_{\rm cut}$. Use the compound-Poisson P(D) with the map mean removed and selection applied. It matches the map ground truth to within $\sim$2 percentage points.
- **Recipe.** (i) Compute $\Sigma_{\rm det}(q_{\rm cut})$ from HMF $\times$ completeness. (ii) Q1: Pois(ΣA), multiplied by $1 + \bar{w}$ if clustering matters. (iii) Q2: build $R(x)$ from the filter cross-response with halos down to $\sim 10^{13}$, FFT to P(D), convolve with the noise, apply the cut and the brightest-contributor rule. (iv) Calibrate absorption against your own peak finder by injecting into maps.